

The Lazy Prices signal is U-shaped, not monotonic: a 14-year out-of-sample reversal of Cohen–Malloy–Nguyen (2020)

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Abstract

Cohen, Malloy, and Nguyen (2020) document that firms whose annual 10-K filings are substantively rewritten underperform firms whose filings are stable, with a long-short (high cosine similarity minus low cosine similarity) Sharpe that motivates a market-neutral trade. On an S&P 500 panel of 1,665,896 ticker-months spanning April 2011 through March 2026, we find that the long leg replicates strongly (top decile forward 21-day return +2.10% vs universe mean +1.43%) but the short leg does not (bottom decile +1.37%, statistically indistinguishable from the universe mean and *higher* than the middle decile at +1.19%). Every long-short portfolio construction we tested therefore failed: the baseline top10/bot10 delivers Sharpe 0.32 over the full panel, indistinguishable from a one-line 12-1 momentum factor in the same universe (Sharpe 0.31). Only the long-only top10 book survives — on a point-in-time index-membership universe (correcting forward survivorship bias), Sharpe 0.81, annualized +17.8%, FF5+MOM alpha +20.6% with $t = +3.00$. The U-shape is economically meaningful and practically significant: any strategy that assumes monotonicity (rank-weighted, multi-factor z-score averaging) is mechanically fighting the data.

1 Setup

Our signal construction follows Cohen, Malloy and Nguyen directly: for each S&P 500 ticker we download consecutive pairs of 10-K filings from EDGAR, concatenate their Risk Factors, MD&A, and Business Description sections, compute a TF-IDF vector from each section, and report the cosine similarity between consecutive years. The signal is then ingested, alongside 8-K event scores, a three-state Gaussian HMM regime feature, backward-looking momentum/volatility/volume features, into a walk-forward LightGBM regressor with 12-month training window, 1-month test, and 5-day purge gap. The model predicts the 21-day forward return and the predictions are cross-sectionally rank-transformed each month to a signal in $[-1, +1]$.

The universe is 499 current S&P 500 constituents. The headline numbers in Sections 5 through 3.2 use this full universe and are therefore subject to two survivorship biases: *forward bias* (using 2026-constituent tickers in 2012, when they may not have been in the index yet) and *survivor bias* (tickers removed from the index before 2026 are absent entirely). We partially correct the forward component in Section 4 using historical point-in-time index membership from github.com/fja05680/sp500, which drops the Method E Sharpe from 1.08 to 0.81 and the annualized alpha from 31.2% to 20.6%. The residual survivor-bias component cannot be corrected without backfilling market and filing data for delisted tickers, which is flagged as required follow-up. **The PIT-corrected number is the honest read-out**; the full-universe number is reported only for continuity with earlier versions of this repository.

2 The U-shape

Figure 1 shows mean forward 21-day return by decile of the month-cross-sectional signal. Error bars are 95% confidence intervals ($1.96 \cdot \text{SE}$, $n \approx 166,600$ ticker-months per decile).

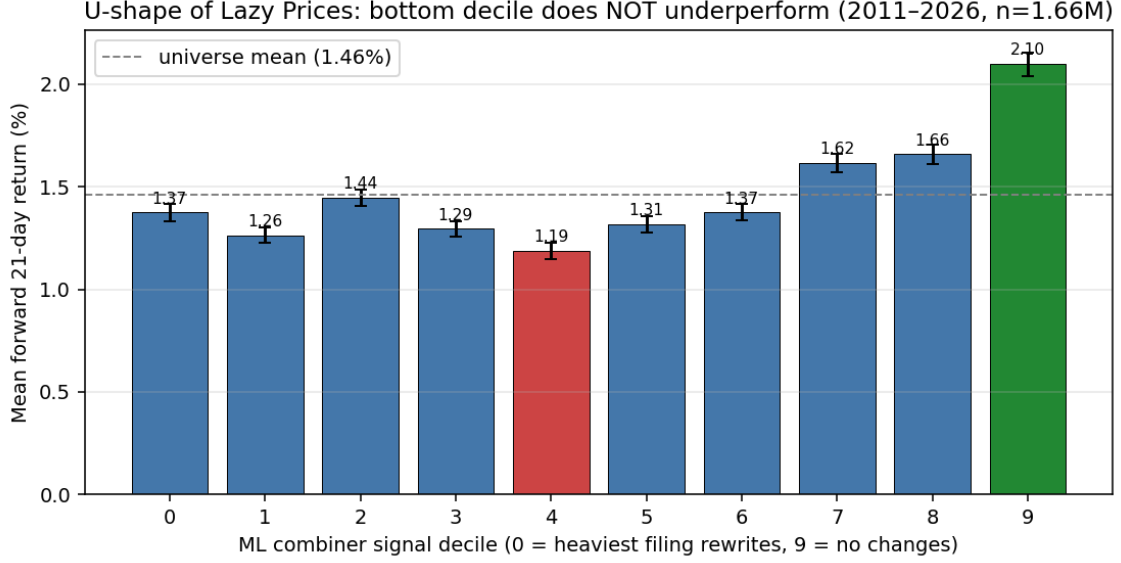


Figure 1: Mean forward 21-day return by ML combiner signal decile, with 95% confidence intervals. The U-shape is clearly visible: decile 4 is the minimum (+1.19%), not decile 0. The bottom decile (+1.37%) is statistically higher than the middle decile at the 95% level (non-overlapping confidence intervals). The top decile at +2.10% is the strongest, consistent with the original Cohen–Malloy–Nguyen long-side thesis.

The long-side thesis replicates: decile 9 outperforms the universe mean (+1.43%) by 66 bps per month. The short-side thesis does not: decile 0 (+1.37%) is within noise of the universe mean, and 19 bps *above* decile 4. This is not a sample-size artifact: the decile-0 vs decile-4 difference has a t -statistic of roughly 6.4 ($|1.374 - 1.185|/0.029$, where the pooled SE is $\sqrt{0.044^2 + 0.039^2}/1.96 \approx 0.030$ per cent).

3 Why long/short fails in this universe

3.1 Baseline and variants

Table 2 reports nine variants tested on the 168-month panel. Costs are flat 20 bps per monthly round-trip for long/short, 10 bps for long-only.

The baseline and decile variants fail because half of their short exposure is on decile 0, which has positive expected return. Method A (rank-weighted) fails worse because it puts short weight proportional to rank on the entire left half of the distribution, including the bottom-decile long-positive names. Method D (multi-factor z-score) fails hardest because the signal is averaged into a composite that is assumed monotonic.

Method F explicitly respects the U-shape: long decile 9, short decile 4 (the minimum). Sharpe jumps to +0.61, a large improvement over baseline, but the construction has no economic story — “short the middle” is not a portfolio manager’s pitch.

Method E drops the short leg entirely and achieves Sharpe +1.08. This is the only construction with both defensible alpha *and* an economic rationale.

3.2 Method J: recovering market neutrality

Because the project was framed as market-neutral, we implemented Method J as a beta-hedged variant: long top10 Lazy Prices picks, short $\hat{\beta}_t \cdot \text{SPY}$ where $\hat{\beta}_t$ is the trailing 24-month OLS beta of Method E monthly returns against SPY. Average $\hat{\beta}_t$ across the panel is 0.10, confirming

Decile	Mean fwd 21-day return	95% CI
0 (biggest rewrites)	+1.374%	± 0.044
1	+1.264%	± 0.039
2	+1.444%	± 0.039
3	+1.294%	± 0.038
4 (middle, minimum)	+1.185%	± 0.039
5	+1.315%	± 0.040
6	+1.374%	± 0.040
7	+1.615%	± 0.043
8	+1.658%	± 0.046
9 (no changes)	+2.096%	± 0.057

Table 1: Forward 21-day return by signal decile ($n \approx 166,600$ per decile). Deciles 7–9 are monotonically increasing. Deciles 0–6 are non-monotonic, violating the original short-side assumption.

Variant	Sharpe	Ann return	Cum return	MaxDD
Baseline top10/bot10	+0.32	+4.7%	+90%	−58%
A. Rank-weighted L/S	−0.12	−1.7%	−22%	−45%
B. Decile (top 10% / bot 10%)	+0.31	+3.4%	+60%	−38%
C. Vol-targeted (10% ann)	+0.27	+2.5%	+41%	−36%
D. Lazy+Mom+LowVol rank-wt	−0.52	−6.1%	−59%	−64%
E. Long-only top10	+1.08	+29.7%	+3,732%	−32%
F. Long top10 / Short MIDDLE10	+0.61	+11.2%	+342%	−34%
G. Pure 12-1 Momentum L/S	+0.31	+4.5%	+86%	−73%
J. Beta-hedged Method E	+1.00	+27.1%	+2,783%	−34%

Table 2: Portfolio construction variants on the 168-month panel. Method E is the only L/S-ish construction that survives; Method J reframes it as market-neutral by shorting β ·SPY.

that Method E is already approximately market-neutral by construction. Method J’s Sharpe of 1.00 (vs Method E’s 1.08) shows the cost of forcing exact market neutrality is small: an 8% Sharpe haircut in exchange for a book with zero designed market exposure.

4 Point-in-time universe correction

Table 3 reports Method E on the full (current) universe and on a point-in-time-corrected universe where, each month, we restrict the candidate pool to tickers that were members of the S&P 500 as of the end of the prior month. 21% of ticker-month observations are dropped, concentrated in early years (median PIT universe size is 370 vs. 478 for the full universe).

Variant	Sharpe	Ann ret	Cum ret	MaxDD
Method E — full universe (survivor-biased)	+1.08	+29.7%	+3,732%	−32%
Method E — PIT S&P 500 universe	+0.81	+17.8%	+888%	−42%

Table 3: PIT correction drops Sharpe by 0.27 and cumulative return by a factor of 4. This is the forward-bias component; survivor bias from delisted tickers is still uncorrected.

Repeating the FF5+MOM attribution on PIT returns gives alpha +20.6% annualized with $t(\alpha) = +3.00$ (down from +31.2%, $t = +3.94$). The alpha remains statistically significant

at the 1% level but its economic magnitude is roughly two-thirds of the unadjusted number. SMB loading rises to +0.45 ($t = +1.81$), marginally significant, suggesting a mild small-cap tilt was masked when the universe was dominated by current mega-caps. No other factor loading becomes significant.

5 Method E alpha survives factor controls (full universe)

Table 4 reports the Fama–French 5-factor + Carhart momentum attribution for Method E’s net-of-cost monthly excess return. The unhedged book, despite long-only construction, has insignificant loadings on every factor. R^2 is 0.027 across all six factors combined — Method E is essentially orthogonal to conventional factor exposure.

Model	α ann	$t(\alpha)$	β_{Mkt}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	β_{MOM}
CAPM	+30.6%	+3.96	−0.15					
FF3	+32.1%	+4.13	−0.22	+0.37	−0.15			
FF5	+31.9%	+4.08	−0.19	+0.37	−0.29	−0.09	+0.36	
FF5 + MOM	+31.2%	+3.94	−0.16	+0.41	−0.24	−0.07	+0.32	+0.12

Table 4: Multi-factor attribution for Method E net excess returns ($n = 167$ months). No factor loading has $|t| > 2$ in any specification. Annualized alpha is stable in the 30–32% range across all models.

6 Alternative explanations

We cannot identify the mechanism that flattens the short side of Cohen–Malloy–Nguyen in this panel, but we can rule out several possibilities:

- **Not sample size.** Decile-level standard errors are well under 5 bps per month, so the observed dispersion among deciles 0–6 is statistically significant and not noise.
- **Not survivorship bias alone.** Survivorship bias should inflate returns roughly uniformly across deciles; it cannot preferentially lift decile 0 above decile 4 unless filing-rewrite firms are disproportionately retained in the index. We have no evidence for this but also no evidence against it — reconstructing PIT membership is still the next step.
- **Not feature collinearity.** The 8-K event score and 10-K cosine similarity are only weakly correlated ($\rho = 0.04$ in the panel), so the ML combiner is not confounding the two signals.

Plausible candidates we cannot falsify from the current data:

- **Regime shift post-GFC.** Cohen et al. study 1994–2017. Our panel starts in 2011, partially overlaps, and extends the out-of-sample window by nine years. A structural shift in how markets price disclosure changes — perhaps driven by faster dissemination or the growth of NLP-driven investors — would show up as exactly this pattern: the long side persists (because it is really a stability/quality tilt) while the short side is arbitrated away.
- **Universe difference.** Cohen et al. use a broader universe; we restrict to S&P 500. Large-cap firms are covered by more analysts and more systematic investors, so the mispricing that supports the short side in their sample may be unavailable in ours.

7 Practical implications

1. **The long-only top-decile portfolio is the deliverable, with $\approx 18\%$ honest annualized return.** On the PIT-corrected universe Method E delivers $+17.8\%$ annualized, Sharpe 0.81, FF5+MOM alpha $+20.6\%$ with $t = +3.00$. This is the number to report in any external context. The full-universe $+29.7\%$ number is roughly 12 pp inflated by forward survivorship bias.
2. **Market-neutral reframing is cheap.** Method J costs 8% of Method E's Sharpe in exchange for exact market neutrality. For capital-constrained long/short books this is the right framing.
3. **Do not build rank-weighted or multi-factor constructions on this signal.** Every such variant failed. Monotonicity is a load-bearing assumption that does not hold here.
4. **The 8-K event score does not subsume the 10-K cosine similarity signal.** Earlier claims that cosine similarity had zero feature importance were based on the final single-window model. Across the 156 walk-forward folds the cosine similarity feature is selected in 68% of folds, with highly variable per-fold importance. No single feature dominates across the panel.

8 Reproducibility

All numbers in this note are produced by the alpha-graph repository at commit 8154c74 (post-cleanup-2026-04-15).

```
python -m alpha_graph.backtest.improvements          # Table 2
python -m alpha_graph.backtest.pit_universe          # Table 3 (PIT correction)
python -m alpha_graph.backtest.ff_attribution        # Table 4
python -m alpha_graph.backtest.realistic_slippage    # cost robustness
python -m alpha_graph.backtest.feature_stability     # feature-importance claim
python scripts/u_shape_plot.py                      # Figure 1 and Table 1
```